Data Analytics Project: IGN Game Reviews Analysis

Dataset Used: ign.csv

This document outlines a structured, end-to-end data analysis project based on game reviews from the IGN dataset. The project is divided into five scenarios, increasing in difficulty from basic data handling to advanced analysis and visualization, ensuring comprehensive skill coverage.

Project Goal: To analyse game review scores, platform popularity, and genre distribution over time using the ign.csv dataset.

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (np.diff).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

Project Coverage

Category	Skills Covered
Data Pre-processing	Data Loading, Missing Value Handling (Mean/Mode), Type Conversion, Column Removal
Pandas Operations	Grouping, Filtering, Aggregation, Feature Engineering (score category, editor's choice conversion)
NumPy Calculations	Array conversion, Yearly score growth calculation (np.diff)
Visualization	Line Chart, Bar Chart, Pie Chart, Histogram, Stacked Bar Chart, Saving Plots

Scenario 1: Basic Data Loading & Cleaning (Easy)

Description:

The ign.csv dataset is loaded using Pandas and explored through head(), tail(), and shape to understand its structure. The unnecessary column "Unnamed: 0" is dropped, and missing values in score, genre, and platform are identified using isnull().sum(). Missing numeric values in score are filled with the mean, while the categorical column genre is filled with the mode to maintain data integrity.

ask	Description
1.	Load the ign.csv dataset using Pandas.
2.	Display the first 5 rows (head()), last 5 rows (tail()), and the shape of the dataset.
3.	Remove the unnecessary column: "Unnamed: 0".
4.	Check for missing values in score, genre, and platform.
5.	Handle missing values: Fill the numeric column score with the mean, and the categorical column genre with the mode.
6.	Ensure correct data types: Convert score → float and release_year, release_month, release_day → integer.

Output

```
   Unnamed: 0 score_phrase                                title  ... release_year release_month  release_day
0           0     Amazing                     LittleBigPlanet PS Vita  ...         2012            9           12
1           1     Amazing  LittleBigPlanet PS Vita -- Marvel Super Hero E...  ...         2012            9           12
2           2       Great                       Splice: Tree of Life  ...         2012            9           12
3           3       Great                                     NHL 13  ...         2012            9           11
4           4       Great                                     NHL 13  ...         2012            9           11
```

```
Displaying Last Five  Rows
      Unnamed: 0 score_phrase                              title  ... release_year release_month  release_day
18620       18620        Good              Tokyo Mirage Sessions #FE  ...         2016            6           29
18621       18621     Amazing       LEGO Star Wars: The Force Awakens  ...         2016            6           29
18622       18622    Mediocre  Star Ocean: Integrity and Faithlessness  ...         2016            6           28
18623       18623  Masterpiece                              Inside  ...         2016            6           28
18624       18624  Masterpiece                              Inside  ...         2016            6           28
```

```
dtype: int64
score_phrase          str
title                 str
url                   str
platform              str
score             float64
genre              object
editors_choice        str
release_year        int64
release_month       int64
release_day         int64
dtype: object
```

Conclusion:

The IGN dataset containing over 18,000 game records was successfully loaded, explored, and cleaned for analysis. The Unnamed: 0" column was removed, and missing values in score and genre were filled using mean and mode imputation respectively**.** Finally, data types are corrected by converting score to float and release_year, release_month, and release_day to integer for accurate analysis.

Scenario 2: Line Graph (Score Trend) + Save (Easy)

Description:
The dataset is grouped by release_year and the average score per year is calculated using Pandas. The resulting values are converted into NumPy arrays for efficient numerical handling. A line graph is then plotted using Matplotlib with proper title, axis labels, and data point annotations.

Task	Description
1.	Group data by release_year.
2.	Calculate the average score per year using Pandas.
3.	Convert the resulting average scores and years into NumPy arrays.
4.	Plot a line graph with release_year on the X-axis and average score on the Y-axis.
5.	Add the title: "Average Game Score Over Years" and proper axis labels.
6.	Save the graph: plt.savefig("avg_score_trend.png").

Graph

The graph shows average game scores from 1970 to 2017, showing a sharp decline with highest of 8.5 in 1970, down to a lowest in 1995. After hitting that bottom, scores gradually recovered through the 2000s, stabilizing between 7.0 and 7.6 in recent years. Overall, the trend suggests that while early games were rated highly, modern games have settled into a consistent mid-range score after years of ups-downs.

Conclusion:

The analysis of average game scores over the years reveals an interesting trend — scores peaked in the early 1970s, experienced a significant decline through the 1990s, and then gradually recovered in the 2000s onward. The steady improvement in recent years suggests that the gaming industry has matured, with more consistently rated titles being released. Overall, while scores haven't returned to their 1970s highs, the upward momentum in the 2010s indicates a positive direction in game quality and reception and finally, the chart is saved as avg_score_trend.png to visualize the trend of average game scores over time.

Scenario 3: Filtering + Bar Chart + Save (Medium)

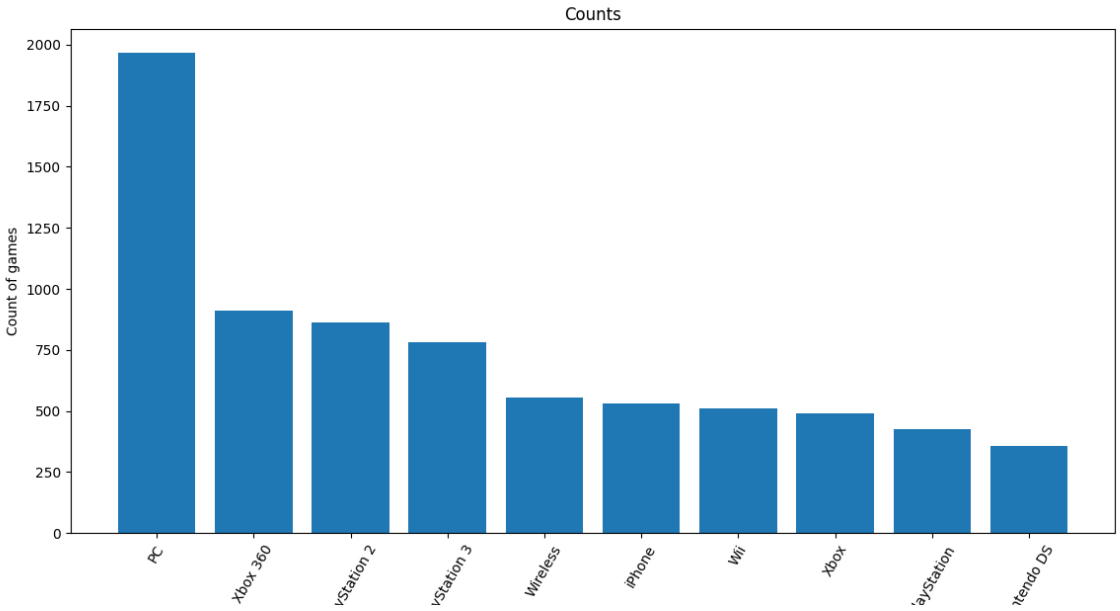
Description:

This scenario focuses on filtering and visualizing high-rated games from the dataset using Python. Games with a score greater than 7 are extracted to focus only on top-quality titles. The filtered data is then grouped by platform to count how many high-rated games each platform has. The top 10 platforms are selected using Pandas and converted into NumPy arrays for plotting. Finally, a bar chart is created and saved as top_platforms_bar.png to clearly compare high-rated game counts across platforms.

Task	Description
1.	Filter the dataset to include only games where score > 7.
2.	Count the number of high-rated games per platform.
3.	Select the top 10 platforms using Pandas.
4.	Convert the platform counts into NumPy arrays.
5.	Plot a bar chart with platform on the X-axis and count of games on the Y-axis.
6.	Rotate x-axis labels for readability.
7.	Save the graph: plt.savefig("top_platforms_bar.png").

Graph:

This bar chart displays the Count of High-Rated Games across the Top 10 Platforms. PC dominates by a large margin with nearly 2,000 high-rated games, far ahead of all other platforms. Xbox 360 and PlayStation 2 follow closely in second and third place with around 900 and 860 games respectively. PlayStation 3, Wireless, iPhone, Wii, Xbox, PlayStation, and Nintendo DS round out the remaining spots, each ranging between 350 and 780 games. The chart clearly shows that PC is the leading platform for high-rated games, while handheld and mobile platforms like Nintendo DS and iPhone rank toward the lower end of



Conclusion:

The bar chart clearly shows that PC dominates all other platforms with nearly 2,000 high-rated games, far ahead of its closest competitors Xbox 360 and PlayStation 2. Traditional consoles like PlayStation 3, Wii, and Xbox follow steadily, reflecting their strong popularity during the peak gaming era of the 2000s. The presence of mobile platforms like iPhone and Wireless in the top 10 highlights the growing influence of mobile gaming in the industry. Overall, the data confirms that PC and legacy consoles lead in high-rated game libraries, while mobile platforms are gradually emerging as competitive alternatives.

Scenario 4: Aggregation + Pie Chart + Save (Medium)

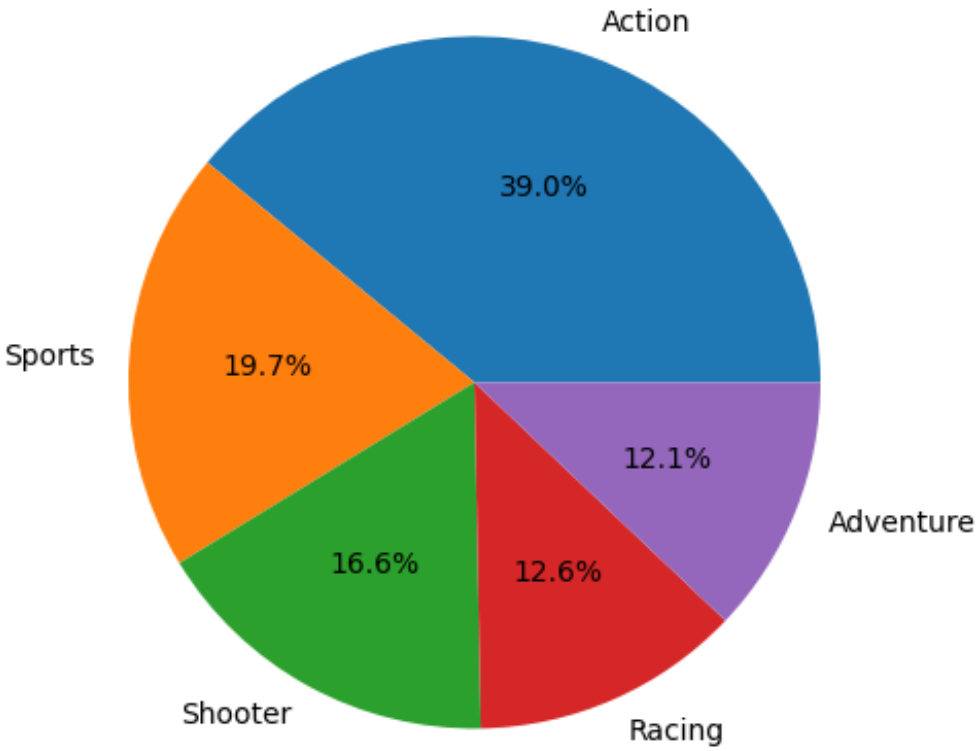
Description

This scenario focuses on analysing the distribution of games across different genres using aggregation and pie chart visualization. The dataset is grouped by genre and the total count of games per genre is calculated using Pandas. The top 5 genres are selected to keep the chart clean and focused on the most dominant categories. Labels and values are prepared and passed into a pie chart with percentage labels using autopct for clear proportional representation.

Task	Description
1.	Count the total number of games per genre.
2.	Select the top 5 genres using Pandas.
3.	Prepare labels (genres) and values (counts) for the pie chart.
4.	Plot a pie chart using genre as labels and count as values.
5.	Add percentage labels using autopct (e.g., '%.1f%%').
6.	Save the graph: plt.savefig("genre_distribution.png").

Graph:

The pie chart shows the distribution of games across the top 5 genres. Action dominates with the largest share at 39.0%, making it nearly double the size of the next genre. Sports follows in second place at 19.7%, then Shooter, Racing, and Adventure hold small shares of 16.6%, 12.6%, and 12.1% . Overall, Action is the most popular genre by a significant margin, while the remaining four genres share a more balanced portion of the distribution.



Conclusion:

The pie chart confirms that Action is the most dominant genre, with 39% of all games and two times the share of Sports at 19.7%. Shooter, Racing, and Adventure have similar and smaller portions, ranging between 12% and 17%, indicating a fairly even spread among the remaining genres. Overall, the data suggests that Action games lead the industry by a wide margin, han others

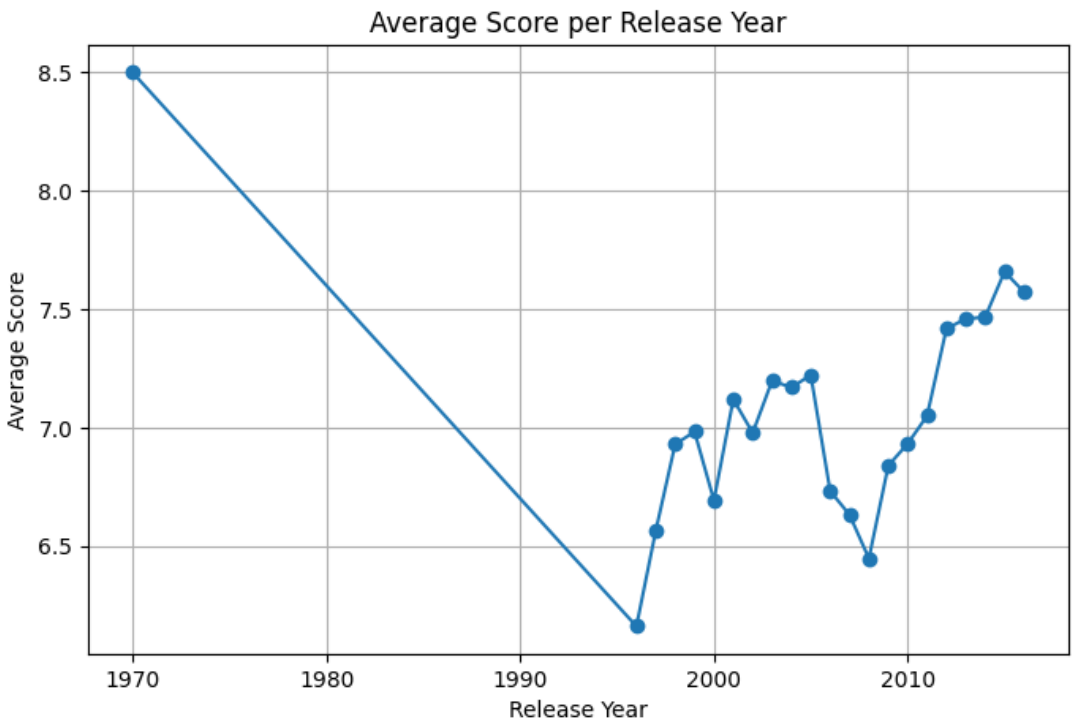
Scenario 5: Advanced Analysis + Multiple Graphs (Hard)

Description:
This advanced scenario performs multi-dimensional analysis on the games dataset by combining data transformation, NumPy computations, and multiple visualizations. A new score_category column is created to classify games as Excellent, Good, or Average, and the editors_choice column is encoded numerically for analysis. NumPy's np.diff() is used to calculate year-over-year score growth trends..

Task	Description
1.	Create a new column score_category: "Excellent" for score ≥ 9, "Good" for $7 \le$ score < 9, and "Average" for score < 7.
2.	Convert the editors_choice column: 'Y' $\rightarrow 1$, 'N' $\rightarrow 0$.
3.	Use NumPy's np.diff() to calculate the **yearly score growth** based on the average yearly scores from Scenario 2.
4.	Plot a **Line Graph** showing the trend of average score per release_year.
5.	Plot a **Stacked Bar Chart** showing the count of score_category per release_year.
6.	Plot a **Histogram** showing the distribution of score.
7.	Save the graphs: plt.savefig("score_trend.png"), plt.savefig("score_category_stacked.png"), plt.savefig("score_distribution.png").
8.	Identify: Which years had the highest scores, whether high scores increased over time, and if editors_choice correlates with high scores.

Graph (Optional):

This line graph displays the average score per release year spanning several decades from the 1970s to the mid-2010s. Scores started at a notably high point in the early 1970s, then experienced a long and decline reaching the lowest point in the mid-1990s. After that low, scores began a gradual recovery through the 2000s with visible fluctuations and minor dips along the way.



Conclusion:

The graph reveals a dramatic decline in average game scores from the 1970s to the mid-1990s, followed by a steady and consistent recovery through the 2000s and 2010s. This trend suggests that while the gaming industry went through a turbulent period of low-quality releases, it has since matured and improved significantly in delivering higher-rated games

Final Conclusion:

This project demonstrated the power of Pandas, NumPy, and Matplotlib in analysing and visualizing a real-world gaming dataset across multiple scenarios. Some things revealed that PC dominates in high-rated game count, Action is the most popular genre, and average game scores showed a long-term recovery after a significant decline in the mid-1990s. Overall, this project highlighted how structured data analysis and visualization can effectively transform raw data into clear and actionable insights.

What you have learned

Through this project, I learned how to use Pandas for data filtering, grouping, aggregation, and column transformations like creating classification columns. I understood how NumPy arrays handle numerical data efficiently, including using np.diff() to calculate year-over-year growth trends. I gained experience building various chart types with. Most importantly, this project taught me how to connect data preparation, analysis, and visualization into a complete end-to-end workflow that produces meaningful conclusions from raw data.